

Positive-Case Count and Synthetic Data Fidelity: A CART-Based Analysis of MIMIC-IV Clinical Data

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Abstract

A logistic fit’s precision is set by the number of positive cases rather than by the sample size. This report asks whether the same threshold governs a generator. CART-based synthesis (**synthpop**) is run on MIMIC-IV ($n = 70,954$ ICU stays) under two sweeps that hold the generator fixed and vary only the input: one raises the rare stratum’s positive count k at fixed sample size, the other raises the sample size n at fixed k . Together they separate count from prevalence.

The threshold does carry over, but the generator adds none of its own. Raising k drives divergence to zero between counts of forty and eighty, and over a matched eightfold increase it cuts interquartile width by 42 to 66% in all six series, while the same increase in n moves neither. Neither the real arm’s divergence rate nor a logistic fit to the same subsamples involves the synthesizer at all, so the threshold belongs to the stratum and **synthpop** inherits it rather than imposing a second one. What synthesis contributes instead is a bias no count removes: in a companion simulation with a known data-generating process, the synthetic median recovers 99% of a true interaction of 0.5 but only 41% of one of 3. The finding transfers as a mechanism and not as a remedy.

A 500-row subsample carries 6 to 7% power against the interactions MIMIC-IV offers, and the smallest one it could detect is seven to twelve times the one actually present. Two recommendations follow: establish that a target effect is estimable in the source data before attributing any discrepancy to the generator, and calibrate variable-selection diagnostics against a permutation null. That null moves from 0.00 to 0.32 with class balance alone.

1 Introduction

That a logistic fit’s precision is set by the event count is established [1, 2], and separation at thin counts has been characterised since [3]. In earlier coursework I found the same under Bayesian logistic regression on the **birthwt** dataset: posterior stability depends on a predictor’s positive-case count rather than on overall sample size. What is not established is whether the same threshold governs a *generator*. Whether a CART-based synthesizer preserves a rare-stratum interaction is a question about which variables a tree spends a split on, not about the curvature of a likelihood, and the two need not share a threshold.

Synthetic data is proposed as a way to share clinical records without exposing patients, and is usually judged by general utility, how closely the synthetic joint distribution resembles the real one [9]. A generator can score well on that and still lose the one coefficient a study was designed around, so what follows holds the generator fixed and measures a single interaction rather than overall similarity. The question matters because of a specific misreading. A researcher handed a synthetic release of MIMIC-IV, finding the CHF by creatinine interaction absent from it, would reasonably conclude that the synthesizer had destroyed a clinical interaction. In the 500 rows in front of them that effect was never detectable to begin with, and the same discrepancy would have appeared with no synthesizer involved at all. Code is available at [GitHub Repository](#).

2 Data

2.1 Cohort

The cohort is MIMIC-IV v2.2 [8], restricted to hospital admissions with an associated ICU stay, one row per stay. Dropping the 3% of stays with no first-day creatinine leaves a pool of 70,954 from which all subsampling draws. In-hospital mortality in the pool is 10.9%, so a 500-row draw carries roughly 54 deaths against the five parameters each fit estimates.

Two features bound what the pool can serve as a reference for, and both land on the full-pool estimate rather than on the sweeps. The dropped stays are not missing at random, being far shorter and more than twice as often fatal, so the reference is the interaction among stays that survived long enough to be measured. And rows repeat patients, so full-pool standard errors are clustered (Table 2). The sweep comparison is untouched by both, drawing its two arms from the same pool.

2.2 Variables

Three predictors enter every fit: age, first-day maximum creatinine, and one comorbidity flag. Comorbidities are simplified ICD-9/ICD-10 prefix flags rather than a validated index such as Elixhauser or Charlson, since the sweeps need a binary flag with a known positive count. The three least prevalent of the five screened were carried forward, COPD, CHF and Diabetes, running 11 to 30% positive in the pool; even the rarest is too common to supply a thin stratum on its own, so the rare stratum is manufactured by subsampling rather than found in the data. Creatinine is capped at the 99.5th percentile and left untransformed, so every creatinine coefficient below is per mg/dL. The notebook carries the screening table and the variable distributions.

3 Methods

3.1 Experimental Design

Both sweeps draw stratified subsamples, synthesize each with `synthpop::syn()` [7], and fit

$$\text{hospital_expire_flag} \sim \text{age} + \text{creatinine} + \text{comorbidity} + \text{comorbidity}:\text{creatinine}$$

to both the real subsample and the synthetic table, giving paired estimates $\hat{\beta}_{\text{real}}$ and $\hat{\beta}_{\text{syn}}$ of the interaction, at $M = 100$ Monte Carlo replicates per grid point. The size sweep holds the count at 20, the last value at which it still binds, so that if sample size can substitute for count this is where it would show (Table 1).

Sweep	Held fixed	Varied	What moves
Count sweep	$n = 500$	$k \in \{5, 10, 20, 40, 80, 150, 250\}$	count and prevalence together
Size sweep	$k = 20$	$n \in \{250, 500, 1000, 2000, 4000\}$	prevalence only, 8% to 0.5%

Table 1: The two sweeps. Each holds the generator fixed and varies one input.

Throughout, a *series* is one comorbidity in one arm, six in all, and a *cell* is one series at one grid point.

3.2 Analyses

Two analyses run on every grid point. The first compares $\hat{\beta}_{\text{real}}$ against $\hat{\beta}_{\text{syn}}$ on the median and interquartile width of the converged estimates, with the divergence and inestimable rates kept separate; Section 3.3 defines the two ways a fit fails. Throughout, the synthetic arm is fitted the way a data recipient would fit it: one synthetic copy, an ordinary GLM.

The second is a tree diagnosis. It refits the subsamples both sweeps draw with `rpart`, under the control parameters `synthpop::syn.cart` uses, and records whether the comorbidity earns a split. That is read

against a likelihood-ratio test on the same rows, dropping the comorbidity and its interaction together, so both are asked the same question. Every configuration is re-run with the comorbidity column shuffled, which leaves the positive count intact and destroys its association with the outcome, supplying the null.

3.3 Two Failure Modes

Fits at the smallest counts fail often enough that the rates are reported as a result rather than used as a filtering step. A fit is **inestimable** when the table holds no positives and the interaction returns NA; **divergent** when the term is estimated but runs past $|2|$, forty times the full-pool estimate, marking a likelihood maximized at the boundary [5]. Only divergence appears in the real arm, which holds exactly k positives by construction; the synthetic table generates its own column and can produce none.

Rates are shares of all M replicates, so converged, inestimable and divergent sum to one; every quantile below uses converged fits only.

3.4 The Full-Pool Reference

MIMIC-IV carries no known ground truth, so the interaction fit on the whole pool stands in as the reference, $\hat{\beta}_{\text{full}}$, and every subsample estimate is read against it (Table 2). The three interactions differ in sign, CHF and Diabetes dampening creatinine’s main effect while COPD steepens it, and all three are roughly three to four times smaller than that main effect.

Comorbidity	Creatinine	$\hat{\beta}_{\text{full}}$	SE	p
CHF	0.208	−0.0497	0.0141	4.4×10^{-4}
Diabetes	0.234	−0.0709	0.0145	1.1×10^{-6}
COPD	0.188	+0.0668	0.0232	4.0×10^{-3}

Table 2: Comorbidity by creatinine interaction on the full pool. Standard errors clustered on patient.

3.5 What This Design Can and Cannot Resolve

The reference also bounds what the subsamples can show. At $n = 500$ the smallest interaction detectable at 80% power is 0.47 to 0.78, against the 0.050 to 0.071 the pool actually holds, seven to twelve times too coarse; reaching those effects would need roughly 23,000 to 67,000 rows.

No subsample of the size studied here can therefore say whether synthesis recovers a true interaction. That is a trade rather than a concession. What the two sweeps measure instead is precision and failure behaviour, how wide the estimates run and how often a fit cannot be produced at all, and both are readable at any effect size, including one too small to detect. Every claim about bias rests on the simulation of Section 4.3, where the truth is known and large enough to reach.

4 Results

4.1 Count, Not Sample Size, Removes Divergence

Count sweep. Divergence runs at 18 to 40% at $k = 5$ and every series is at zero by $k = 80$ (Figure 1, left), though not at one shared count: two are already there at $k = 40$, and a zero over 100 replicates bounds a rate at 3.6% rather than establishing it as nil, so the threshold sits in $(40, 80]$ for each series. Neither prevalence nor the sign of the interaction moves it.

Size sweep. No sample size does the same. Over a matched eightfold increase in n no series comes near zero (Figure 1, right). What separates cleanly here is the arm rather than the size: the synthetic rate is the higher of the two in fourteen of the fifteen comorbidity-by-size comparisons, and no sample size closes the gap.

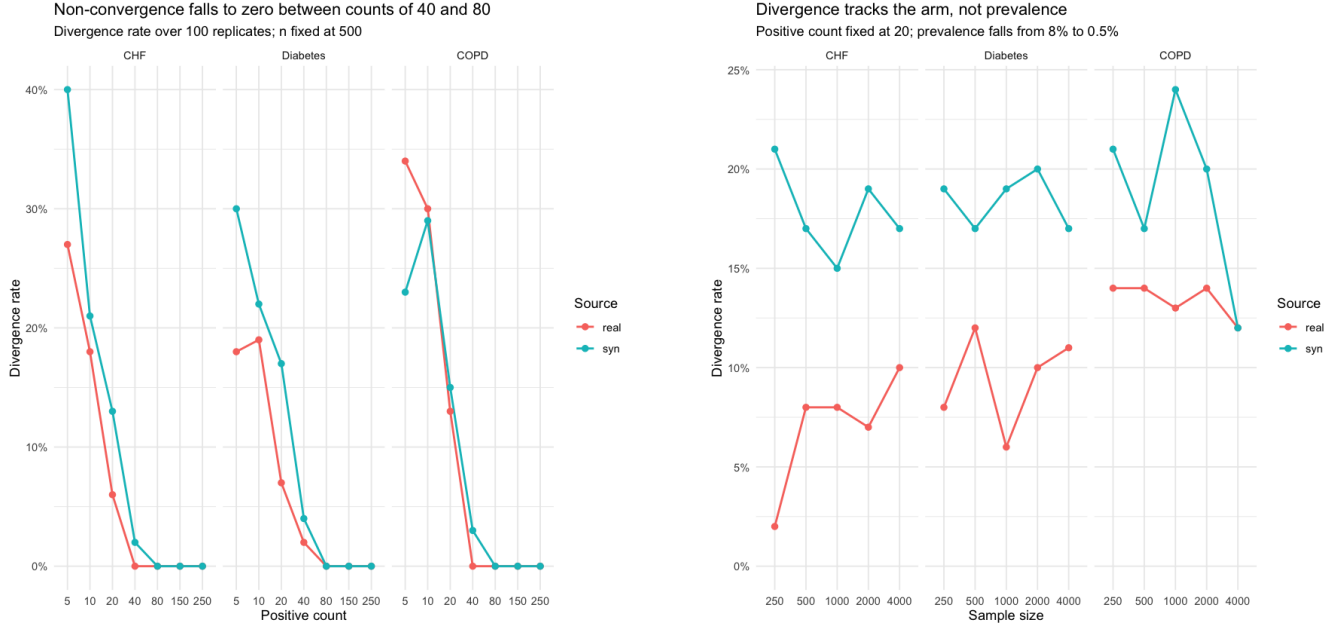


Figure 1: Divergence falls away under the count (left) and not under the sample size (right); the two smallest counts are detection floors.

Detection floor. The $|2|$ cutoff understates divergence at the smallest counts. A penalized fit [4] finds 64 to 76% of real fits separated at $k = 5$ where $|2|$ flags 18 to 34%, because `glm` stops after twenty-five iterations rather than running to the boundary. The two agree from $k = 20$ up, and the penalized fit removes the divergence without moving the count at which it stops. What the cutoff mismeasures is how bad the smallest counts are, not where failure ends.

4.2 The Count Narrows the Spread; the Sample Size Does Not

Count sweep. Spread narrows steadily with the count, every series losing 55 to 69% of its width from $k = 20$ to $k = 250$. It is still not enough: the narrowest cell has an interquartile width of 0.194 against its reference of 0.0497 (Table 2), nearly four times the quantity being estimated. What low counts cost is precision and not location, the middle half of the estimates still containing $\hat{\beta}_{\text{full}}$ in 39 of the 42 cells (Figure 2, left).

Size sweep. Over the matched eightfold window, k from 20 to 150 against n from 500 to 4000, the count cuts width by 42 to 66% in all six series while the sample size changes it by between -19% and $+10\%$ with no consistent sign (Figure 2, right); the count's figure falls below the 55 to 69% above only because this window stops at $k = 150$. Across the full sixteen-fold range of n the widths never leave the band the count sweep recorded at $k = 20$.



Figure 2: The count narrows the spread (left) and the sample size does not (right), on a common scale. Distance from $\hat{\beta}_{full}$; band is the middle 50% of converged fits.

What the count rate explains. An interaction's variance scales as $1/k + 1/(n - k)$, and that rate accounts for both panels. Rescaling the $k = 250$ width by it tracks the real arm at 0.97 across the count sweep; with k held at 20 the $1/k$ term cannot move, so it predicts widths within 4% across the whole size range, against observed scatter several times that in both directions. What it does not account for is the synthetic arm, which sits below the real one and spreads wider than its own count predicts: the same rescaling puts it at 1.14, higher than the real arm at every one of the nine matched cells (Figure 3; sign test $p = 0.004$).

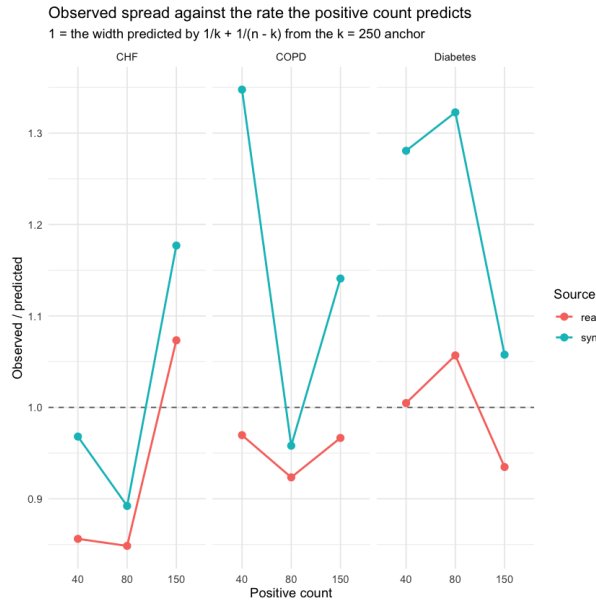


Figure 3: The real arm tracks the width its count predicts; the synthetic arm sits above it at every count.

A sign that sample size does not repair. Width is not the only thing sample size fails to fix. COPD is the one comorbidity whose interaction is positive, +0.0668, and the count sweep recovers that sign once

k passes 20. Holding the count at 20 does not: the synthetic median is negative at all five sample sizes, from -0.077 to -0.159 , matching the -0.177 recorded by the count sweep at the same count. Sixteen times the data leaves the sign where twenty positives put it.

4.3 What No Count Fixes: Attenuation Under a Known Truth

As Section 3.5 set out, MIMIC-IV offers no truth to compare against and no power to reach one. A companion simulation supplies both: a population is generated from a logistic model with a known `chf` $\times$ lactate interaction, and the same count sweep is run at three true values, 200 replicates per cell.

True value	Median estimate		Median error	
	Real	Synthetic	Real	Synthetic
0.5	0.521	0.496	0.164	0.219
1.5	1.73	0.891	0.292	0.713
3.0	2.75	1.22	0.688	1.90

Table 3: Estimates at $k = 250$ against the true value. The last two columns are the median of $|\hat{\beta} - \text{true}|$ over replicates.

Synthesis loses more of the interaction the larger it is. At a true value of 3 the real arm recovers 2.75 and the synthetic arm 1.22 (Table 3), and no count on the grid closes that gap (Figure 4). The row at 0.5 is not the exception it looks like: an estimator that pulls toward zero is accurate against any target near zero, and the error columns show the difference the medians hide, 34% larger on the synthetic side where the two medians agree to within 1%.

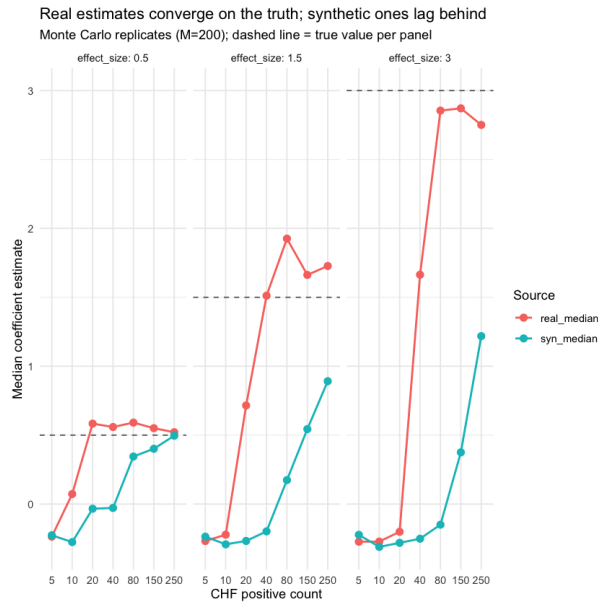


Figure 4: The real arm converges on the truth; the synthetic arm lags at every count. Dashed line is the true value.

At a true value of 3 the real arm diverges in 21 to 29% of replicates while the synthetic arm diverges in 2.5 to 10.5%, the reverse of the MIMIC-IV pattern. A strong interaction puts the real subsample close to separation; the tree smooths that away, so the synthetic fit converges more readily and converges on the wrong value. The attenuation is bought, not incurred.

4.4 The Split Diagnostic Is Unreadable Without Its Null

What looks like recovery in the count sweep is class balance, not signal. The diagnostic rises from 0 at $k = 5$ to 0.26 to 0.40 at $k = 250$, but the permutation null rises with it, to as high as 0.32 (Figure 5, left), since what grows alongside the count is the balance of the column itself, from 1% positive to 50%, and a balanced binary variable is easy to split on whether or not it carries signal [10]. Read across the grid rather than cell by cell the signal is still there. Pooling the grid with a conditional logistic model, `used ~ observed + observed:log2k_c + strata(pair)`, stratified on (comorbidity, replicate) pairs with $\log_2 k$ centred at $k = 40$, puts the real column ahead of the shuffled one by an odds ratio of 2.0 at $k = 40$ ($p = 6 \times 10^{-6}$), narrowing by 0.81 per doubling of the count ($p = 0.015$) as the null closes in.

Holding the count at 20 runs class balance the other way, down to 0.5% positive at $n = 4000$. The null falls with it to near zero while the observed rate stays where the count sweep left it (Figure 5, right), and with the null out of the way the signal is legible without pooling: the real column wins 104 discordant replicates to 32. The diagnostic is readable where its null is near zero, which on this data means low prevalence.

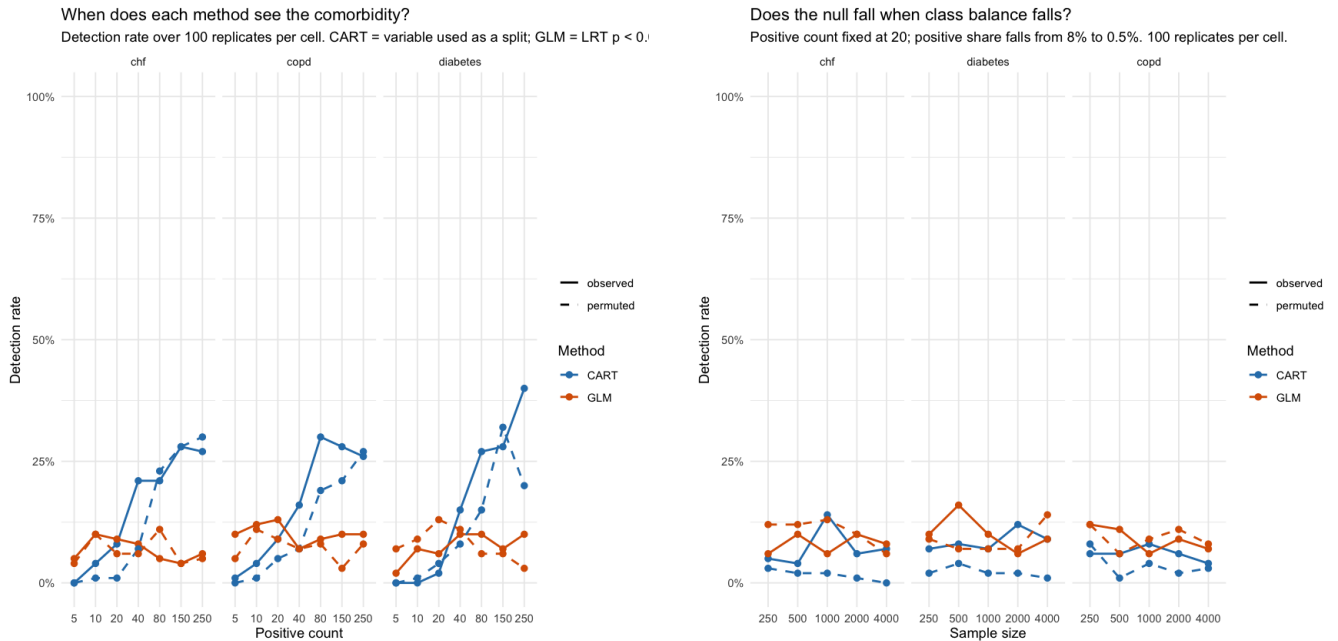


Figure 5: The null climbs with the count (left), so the observed rise is not recovery; with the count held at 20 it falls to near zero while the observed rate stays put (right), leaving the gap legible. Dashed is the shuffled column.

5 Discussion

The design, not the generator. The floor set out in Section 3.5 is what the sweeps returned. Theory gives a 500-row subsample 6 to 7% power against these interactions and the GLM arm returned 6.7 to 10.1%, both within a few points of the 5% a powerless test would produce; the two landing together rules out a defect in the pipeline and locates the failure in the design. Nor does either sweep implicate CART on its own, since the GLM does no better on real labels than on shuffled ones; CART is only the method that fails visibly, by declining to spend a split.

A threshold, but not a cutoff. The count at which divergence stops is not one number: it sits in $(40, 80]$ for each series but lands at different points inside that window. That is the expected shape rather than a ragged result, since van Smeden et al. find no rationale for a fixed events-per-variable criterion [2]. What transfers to a generator is a dependence on the count, not a universal cutoff.

A privacy-fidelity tension. CART’s reluctance to isolate a subgroup with few supporting cases is not simply a limitation to work around. A tree willing to carve out a node of five patients would also be likelier to overfit to, and risk re-identifying, them, and that same reluctance leaves the stratum missing from the synthetic table at $k = 5$. The tension is a property of the method, not an implementation defect, though only its fidelity side is measured here.

Limitations. The diagnostic tree is a proxy for the generator: three predictors rather than the full feature set `synthpop` synthesizes jointly, matched on control parameters but not on its sequential visit order. Both are permissive, which makes non-selection the more reliable outcome. Monte Carlo error at $M = 100$ reaches 13% on interquartile widths, so a cell that misses significance is not evidence that nothing is there. The single-copy GLM on the synthetic side carries no correction for the variance synthesis contributes; proper inference uses several copies and a combining rule [6], so the two arms are compared as procedures in use rather than as matched estimators. One continuous variable and one outcome were examined, two of the five screened comorbidities were left unanalysed, the pool excludes the short, high-mortality stays that carry no first-day creatinine, and all results come from a single MIMIC-IV snapshot.

6 Conclusion

The positive count governs every quantity the two sweeps measure and sample size governs none: divergence falls to zero between counts of forty and eighty, interquartile width falls by 42 to 66% over a matched increase in the count, and neither CART’s use of the comorbidity nor a logistic fit to the same subsamples moves with n . These are four readings on the same replicates rather than four independent confirmations, but two of them come from data the synthesizer never touched. In that narrow sense the finding from Bayesian logistic regression does extend to CART-based synthesis.

What does not extend is a remedy. The generator adds an attenuation no count removes, and the simulation shows it partial even where the target is large enough to reach. At clinical effect sizes neither arm recovers the MIMIC-IV interactions at all, the narrowest cell still spanning nearly four times the quantity being estimated.

For synthetic data evaluation, two things follow: establish that the target effect is estimable in the source data before attributing a discrepancy to the generator, and calibrate any variable-selection diagnostic against a permutation null. Neither is specific to `synthpop`: any generator that declines to model a subgroup it has too few cases to support should show the same threshold, and the two sweeps needed to find it require nothing beyond the source data and the generator itself.

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